Statistical Physics for Machine Learning Exercise 7

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Abstract

We compare five learning rules (Randomized Perceptron, Hebb, Pseudo Inverse, Adaline, and Bayes) in a teacher-student perceptron problem with $N = 20$ and bipolar inputs $\{-1, +1\}$. For each $\alpha = P/N$, we estimate the average generalization error over repeated random teachers/datasets, then analyze the full curve together with low-α and high-α log-log regimes.

Comparison of all learning rules

This experiment is designed as a unified benchmark of all implemented rules under the same data-generation process and evaluation protocol.

Experimental setup

We use a standard teacher-student perceptron setting:

- Input dimension $N = 20$.
- Inputs are i.i.d. bipolar vectors $\mathbf{x} \in \{-1, +1\}^{20}$.
- A random teacher vector $\mathbf{w}^\star$ assigns labels through $\text{sign}(\mathbf{w}^\star \cdot \mathbf{x})$.
- Training-set size is P, and control parameter is $\alpha = P/N$.
- For each P, we repeat the full procedure over many random realizations and report mean test error on fresh samples.

Implemented rules

- **Randomized Perceptron**: online perceptron-style updates with random presentation order.
- **Hebb**: direct correlation-based weight estimate from labeled data.
- **Pseudo Inverse**: minimum-norm linear solution in the under-parameterized/small-α regime.
- **Adaline**: squared-loss optimization (gradient-based), extending beyond $\alpha < 1$.
- **Bayes**: committee/ensemble-based estimate compared with the theoretical Bayes curve.

Implementation structure

The implementation is organized around reusable components:

- Bipolar random-vector generation.
- Dataset construction, teacher-based labeling, shuffling, and error counting.
- Perceptron forward evaluation and rule-specific training procedures.
- Experiment drivers for Monte Carlo repetitions and averaged metrics.
- Post-processing and plotting routines for full-range and asymptotic comparisons.

How to read the figures

The global trend over the full range of α is shown in Fig.(1). The low-α and high-α asymptotic regimes are shown together in Fig.(2) to make direct visual comparison easier.

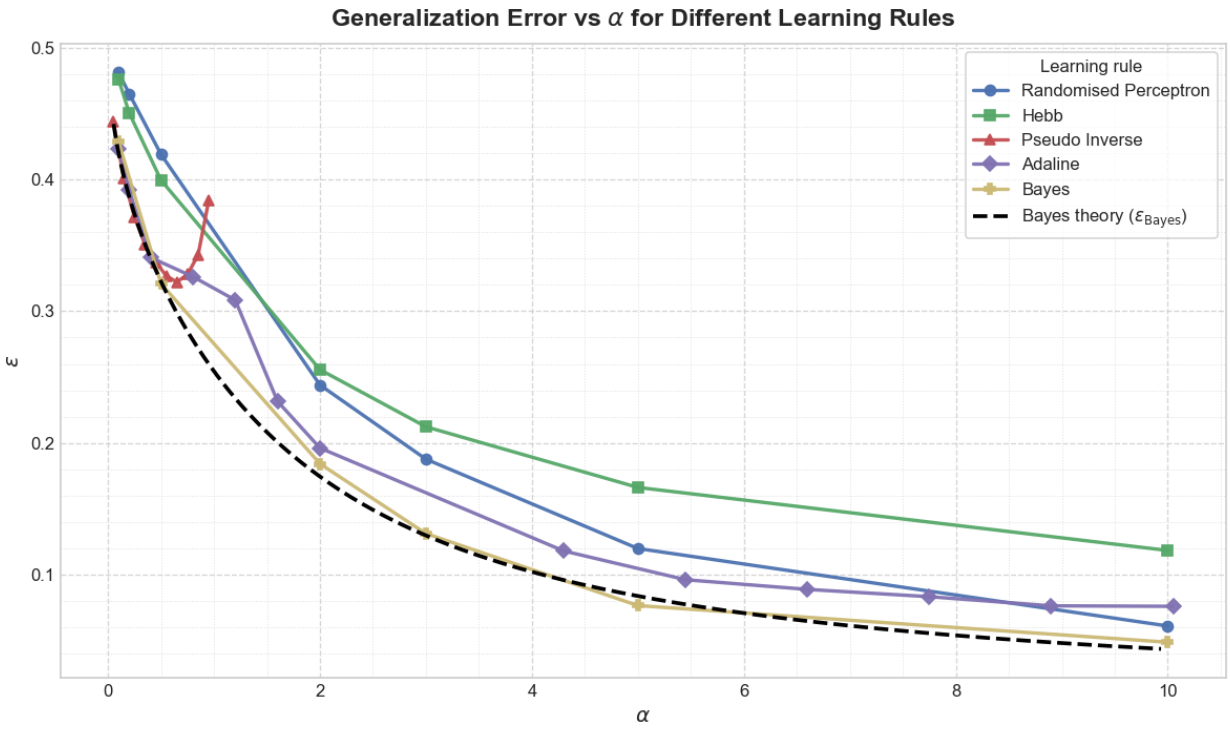
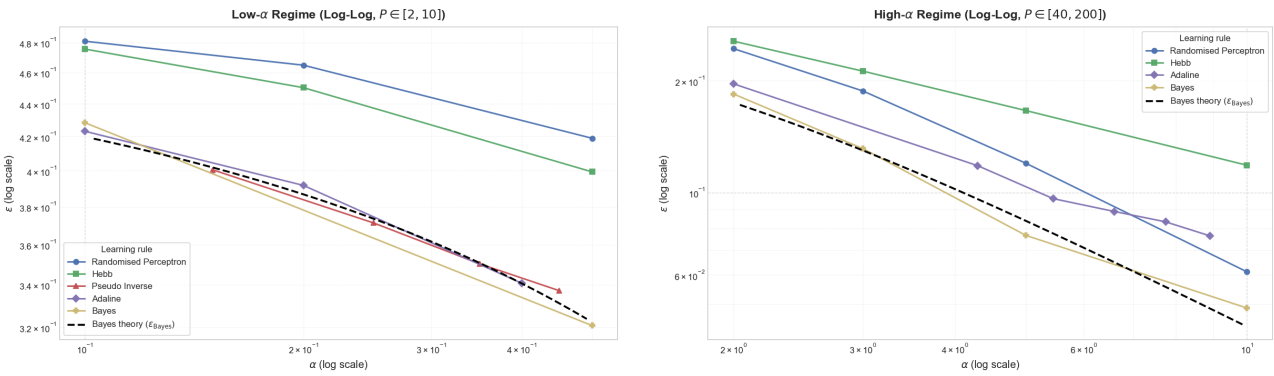


Figure 1: *Generalization error comparison across all learning rules over the full α range.*



(a) Low-α regime (log-log scale). **(b)** High-α regime (log-log scale).

Figure 2: *Asymptotic comparisons for the same set of learning rules: small-sample limit on the left and large-sample limit on the right.*